



Citation (Version of Record):

Pennycook, G., Berinsky, A. J., Bhargava, P., Lin, H., Cole, R., Goldberg, B., Lewandowsky, S., & Rand, D. G. (2024). Inoculation and accuracy prompting increase accuracy discernment in combination but not alone. *Nature Human Behaviour*, 8(12), 2330-2341. <https://doi.org/10.1038/s41562-024-02023-2>

Version:

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Link to published version (if available):

[10.1038/s41562-024-02023-2](https://doi.org/10.1038/s41562-024-02023-2)

Link to publication record on the Bristol Research Portal:

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Technique-based inoculation and accuracy prompts must be combined to increase truth discernment online

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Misinformation remains a serious problem and continues to be a major focus of intervention efforts. Psychological inoculation - a popular intervention approach wherein people are taught to identify manipulation techniques - is being adopted at scale around the globe by technology companies in an effort to combat misinformation. Yet the efficacy of this approach for increasing belief accuracy remains unclear, as prior work has largely focused on technique identification - rather than accuracy judgments - using synthetic materials that do not contain claims of truth or falsity. To address this issue, we conducted 5 studies with 7,286 online participants using a set of news headlines based on real-world false and true content in which we systematically varied the presence or absence of emotional manipulation. Although an emotional manipulation inoculation video did help participants identify emotional manipulation (replicating past work), there was no carry-over effect to improving participants' ability to tell truth from falsehood (i.e. no effect on truth discernment). Encouragingly, however, when the emotional inoculation was paired with an accuracy prompt - i.e., an intervention intended to draw people's attention to the concept of accuracy when they are receiving the inoculation intervention - the combined intervention *did* successfully improve truth discernment by increasing belief in true content. These results generate new insights regarding inoculation, and provide evidence for a key synergy between two popular psychological interventions against misinformation.

Global concerns about the spread of misinformation and its toxic effects on democracy have led to a surge of interest in potential mitigation measures (Athey et al., 2023; Bak-Coleman et al., 2022; Ecker et al., 2022; Kozyreva et al., 2020, 2022; Pennycook & Rand, 2021). One of the most common approaches is to undermine the influence of falsehoods by providing countervailing information. For example, to ensure that people do not believe misinformation (defined here as information that is false or misleading, whatever the intent), research has focused on assessing the efficacy of fact-checks, corrections, and “debunks” (Chan et al., 2017; Lewandowsky et al., 2012; Nieminen & Rapeli, 2019; Porter & Wood, 2021). Although these approaches show promise – corrections do tend to undermine the effect of misinformation and rarely backfire (Wood & Porter, 2019) – such an approach is a loss-mitigation strategy at best. Corrections necessarily occur after misinformation has already spread online, and thus should only be one element of a broader strategy.

Psychological inoculation

Researchers and practitioners have therefore also focused on strategies that attempt to get ahead of misinformation and limit its potential influence before people are exposed. Recent research demonstrates that “prebunking” misinformation by providing corrective information *in anticipation* of falsehoods can protect people against being misled (Cook et al., 2017; Lewandowsky & van der Linden, 2021; Traberg et al., 2023). One promising prebunking strategy that has been gaining popularity is attitudinal or psychological “inoculation”. The goal of inoculation is to provide individuals with knowledge or abilities that help them spot misinformation by exposing them to a weakened form of a misleading technique (or set of techniques) (Roozenbeek et al., 2022; Roozenbeek & van der Linden, 2022; Traberg et al., 2022). Critically, unlike other prebunking techniques, inoculation techniques are not intended to be specific to unique falsehoods, but rather are intended to boost people's general ability to identify misleading techniques that are thought to be common to misinformation (Compton, 2013; Compton et al., 2021)¹.

To illustrate, a recent set of experiments have tested a scalable version of psychological inoculation. Roozenbeek et al. (2022) developed a set of brief videos (less than two minutes each) that teach users about manipulation techniques that are thought to be statistical markers of misinformation (namely emotional language, ad hominem attacks, false dichotomies, incoherence, and scapegoating). To offer an example, the emotional manipulation inoculation video informs users about how emotional language (e.g., words like “disgusting” and “horrifying”) can be used to manipulate users into paying attention to content online (all videos are accessible here, <https://inoculation.science/>). These videos, generally speaking, improved participants’ ability to discern whether there was manipulation in the synthetic tweets that the

¹ Psychological inoculation – and inoculation theory in general – have been applied very broadly in the persuasion literature (Compton, 2013; Compton et al., 2021); however, we focus here on the use of the intervention as a tool to undermine people’s susceptibility to misinformation in particular.

researchers created to either contain the technique or not – with the emotional language inoculation having one of the clearest effects on the ability to discern between manipulative and neutral content.

Two of the inoculation videos were also tested in a quasi-field experiment on YouTube (including the emotional language inoculation that we focus on here). For this, users were presented an inoculation video in the standard course of their YouTube usage as an ad and then were subsequently asked to correctly identify the manipulation technique in a multiple choice question. The researchers found those who viewed the inoculative video were 5-6% more likely to correctly identify a manipulation technique relative to a control group, suggesting that a short video on social media may be sufficient to teach people how to identify misinformation techniques (Roozenbeek et al., 2022). Efforts by technology companies to scale this approach are already underway (Jigsaw, 2023; Klepper, 2023; Mukherjee et al., 2022).

Although inoculation thus shows considerable promise as a scalable approach, research on inoculation has focused on demonstrating people’s ability to detect when they might be misled by boosting their ability to recognize misleading or deceptive argumentation (Roozenbeek et al., 2022; Traber et al., 2023). This is a valuable skill, in particular because it provides protection against poor argumentation irrespective of the specific content. For example, the use of highly emotive or inflammatory language may signal manipulative intent irrespective of whether the message is entirely false, contains seeds of truth, or is accurate.

However, it remains unclear whether inoculation also makes people better at discerning the *accuracy* of claims – that is, whether inoculation increases the ability to differentiate true from false information. Past work has primarily tested the efficacy of inoculation by using content that contains the targeted technique (such as emotional manipulation), but that does not necessarily contain specific truth claims (or claims that are evaluable as true or false). For example, Roozenbeek et al. (2022) used the following as an example of emotional language manipulation: “What this airline did for its passengers will make you tear up - So heartwarming” (with no additional context or information about the airline or what they actually did). Items such as this are intended to be “pure” examples of the technique; naturally, however, the missing context would be present when such interventions are used in the real world and it is important to ascertain whether technique identification skills extend to recognizing the truth or falsehood of statements.

Typically, participants are also explicitly asked about the presence/absence of the technique instead of the accuracy of the statements². Thus, it is unclear if boosting technique recognition carries over to affect judgments about the truth or falsity of true and false content. Does teaching

² One exception to this is Lewandowsky and Yesilada (2021) who measured people’s ability to judge the likely accuracy of problematic content, in addition to examining their detection ability. In this case, inoculation has also shown promise by reducing people’s belief in the reliability of dubious information, although (as in other cases), the test material did not have explicit truth value.

about misleading techniques (on its own) help people detect misinformation? And if not, what can be done to address this issue?

Inattention to accuracy

Notwithstanding the promise of inoculation, there are also reasons to believe that inoculations with a specific focus on misleading techniques (such as using emotional language) may not necessarily carry-over to improve judgments about the accuracy of true or false content. In particular, research has shown that people are often inattentive to accuracy on social media (Arechar et al., 2023; Pennycook et al., 2020, 2021). For example, in many experiments when participants are asked directly to assess whether they believe a set of true and false news headlines, most people are actually quite good at distinguishing between them (i.e., they believe the true news much more than the false news); however, if participants are instead asked to indicate if they would share the news on social media, they are largely insensitive to the veracity of the headlines (Arechar et al., 2023; Pennycook et al., 2020, 2021). In fact, the social media context may itself be responsible for distracting people from thinking about accuracy, as simply having participants make sharing judgments reduces their ability to tell true from false when making accuracy judgments (Epstein et al., 2023).

One approach to overcoming this inattentiveness to accuracy is through very brief interventions that prompt the reader to consider accuracy, known as “accuracy prompts” (or “accuracy primes” or “accuracy nudges”). These prompts have demonstrated promise in improving people’s attentiveness to accuracy when making judgments about what to share online (i.e., by increasing the quality of content that they share) (Epstein et al., 2021; Pennycook et al., 2020, 2021; Roozenbeek et al., 2021; for a meta-analysis see Pennycook & Rand, 2022). This is thought to occur because the prompts redirect participants’ attention to the concept of accuracy, and most participants do not wish to share content that they realize is inaccurate (Arechar et al., 2023; Pennycook et al., 2021). Thus, simply considering accuracy reduces the (undesired) sharing of content which participants would have been able to identify as inaccurate had they thought about it (Lin et al., 2023).

It is therefore possible that the effects of inoculations that focus only on identifying manipulation techniques - but do not address the problem of trying to specifically spot false claims - may not transfer to improved misinformation detection. However, if before delivering the inoculation, the intervention explicitly draws people’s attention to accuracy and the need to differentiate truth from falsehood, this juxtaposition may induce transfer and successfully lead to improved truth discernment. Put differently, there may be a synergy between inoculation approaches (which teach people specific information about manipulation techniques but may not focus their attention on detecting falsehoods) and accuracy prompt approaches (which direct attention to truth and falsity but do not actually teach individuals anything about how to detect falsehoods). Thus, although accuracy prompts are not intended to improve people’s ability to distinguish

between true and false content – as participants are simply having their attention directed to accuracy without being taught anything new – it is possible that inducing an accuracy mindset could improve the effectiveness of psychological inoculations.

Present work

Here we investigate whether psychological inoculation effects carry over to improved truth discernment, as well as the potential synergy between inoculation and accuracy prompting. We show in 3 experiments that, while inoculation reliably boosts people's ability to detect problematic content, it does not by itself increase truth discernment. However, we show in two further experiments that when inoculation is combined with a prompt that reminds people of the problem of misinformation and the importance of accuracy, the combined intervention (but neither prompt nor inoculation in isolation) boosts participants' accuracy discernment.

To assess participants' ability to identify inaccurate claims that systematically vary in the presence of a manipulation technique, we developed a novel set of stimuli that were sourced from real-world true and false news headlines but that were modified to either contain the manipulation technique of emotional language or no such technique (see Figure 1). For this, we took a larger corpus of 83 political headlines (43 false, 40 true) and created two versions for each claim: One intended to elicit strong emotions from the reader by including evocative language and one that did not include an emotional manipulation (and that was also stripped of evocative language). We then took this corpus and completed a pretest where participants ($N = 982$) rated the extent to which a random subset of 15 headlines made them feel emotions (e.g., angry, sad, worried, happy, excited, etc.) using a 6-point scale from 'not at all' to 'extremely'. Participants also rated the headlines in terms of their perceived likelihood and (Democratic v. Republican) partisanship, and indicated whether they would share the headline on social media. From this, we selected 32 headlines where the high emotion version elicited particularly strong emotional reactions relative to the low emotion version (see Table 1). This procedure gave us a set of headlines that not only differed in terms of whether they contained the emotional manipulation technique (as in past work) but that also were pretested to be significantly different in terms of the actual emotions that were elicited by the headlines. Although adding emotional language to the headlines may in theory subtly shift the claims (see Figure 1), here we were able to do so without changing the veracity of the headlines (i.e., while adding emotional language to true headlines may subtly influence their plausibility, they did not become false – and removing emotional language from false headlines did not make them true). Furthermore, we also included a question about perceived plausibility of the headlines in the pretest and the high versus low emotionality headlines did not significantly differ at the item level (see SI)³.

³ It should also be noted that we did not include information about the publisher of the headlines, as they were changed to either have or remove emotional language and thus none of the headlines were, strictly speaking, used by any publishers. In any case, past work has shown that hiding versus revealing publisher information on headlines has little effect on judgments of accuracy (Dias et al., 2020).

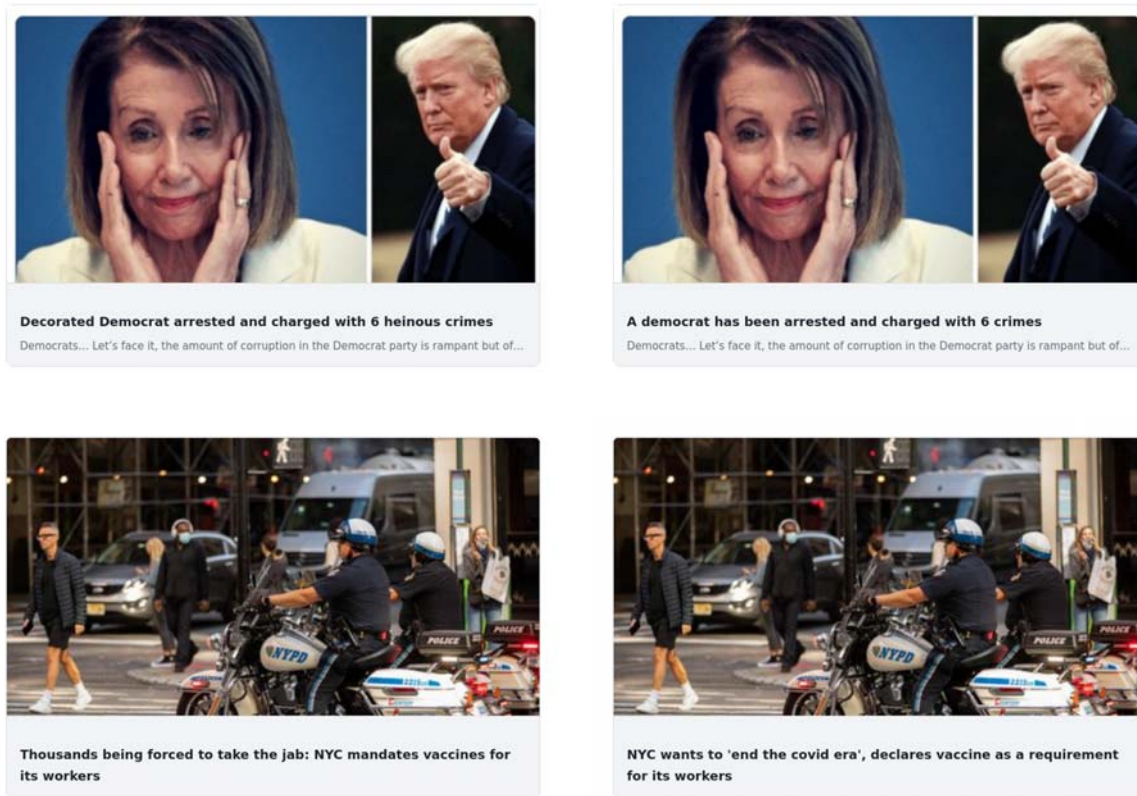


Figure 1. Example false (top) and true (bottom) headlines that were manipulated to contain an emotional language technique (left) or no such technique (right).

Table 1. Average emotionality rating in the pretest for matched high versus low emotionality headlines. Displayed are in-party ratings. I.e., Emotionality ratings among Democrats for Democratic-targeted headlines and among Republicans for Republican-targeted headlines.

	Target	Low Emotionality	High Emotionality	Diff	t-test	Cohen's d
False	Democratic	3.0	3.7	0.7	$t(7) = 19.9, p < .001$	7.02
	Republican	2.8	3.3	0.5	$t(7) = 10.4, p < .001$	3.68
True	Democratic	3.0	3.6	0.6	$t(7) = 6.7, p < .001$	2.37
	Republican	3.1	3.6	0.5	$t(7) = 4.4, p = .003$	1.54

This pretested set of news headlines allows us to test whether the emotional manipulation inoculation is successful in helping people identify misinformation using real-world stimuli that contain true or false claims. In addition, in the studies below participants were asked either about manipulateness or belief (accuracy) separately so that we could investigate if participants spontaneously identify misinformation to a greater extent following the intervention. Importantly, following recent recommendations (Guay et al., 2023), we focus on whether the intervention improves both truth and technique *discernment* (i.e., whether participants judge true headlines as more accurate than false headlines, and whether they judge high emotion headlines as less accurate than low emotion headlines). Researchers have raised concerns that psychological inoculation may increase general skepticism (Modirrousta-Galian & Higham, 2023) and, ultimately, if the inoculation is improving people's ability to *detect* emotional language manipulation, it should result in an improved ability to distinguish between items that do versus do not contain manipulation (for further, see Guay et al., 2023).

Experiment 1

In testing our hypotheses, we focus on whether the inoculation treatment increases *discernment* in two forms. Since we orthogonally manipulate both stimuli facticity and emotionality, we are able to assess two versions of discernment: 1) Do users distinguish between true and false content; i.e., are ratings of belief (manipulateness) lower (higher) for false relative to true news headlines?, and 2) Do users distinguish between emotionally manipulative (high emotion) and not-emotionally manipulative (low emotion) content; i.e., are ratings of belief (manipulateness) lower (higher) for manipulative relative to not-manipulative news headlines?

Method

Participants

Participants were recruited from Amazon's Mechanical Turk via Cloudresearch, which has been shown to provide high quality participants (Douglas et al., 2023). Participants were randomized into an emotional inoculation treatment condition or a control condition, and participants in the control condition were further randomized to one of three control videos. In total, 1,268 participants entered the experiment⁴. However, 33 participants indicated not having social media and did not therefore qualify for the study and 6 participants failed an attention check at the beginning of the study and did not continue. Finally, 3 participants were removed from the survey because they indicated not being able to turn on their volume (which was critical for watching the videos) and 18 participants quit the survey prior to the intervention. A further 178 participants dropped out and did not contribute data to the primary task. The fraction of

⁴ For each study, we performed a sensitivity analysis through simulations to estimate the smallest effect size (Cohen's d) we could detect with at least 80% statistical power, given the mixed-factorial design and sample size. Across the five studies, the smallest effect size we could detect was $d = \sim 0.10$ (d for the five studies: 0.12, 0.13, 0.11, 0.11, 0.10).

participants dropping out did not vary significantly across conditions (15.2% in control, 14.2% in treatment; $\chi^2(1) = 0.23, p = .630$). We therefore had a final sample size of 1,030 participants ($M_{age} = 36$), of whom 364 were male, 610 were female, 30 chose some other response (e.g., trans/non-binary), and 26 did not respond to the gender question.

Materials

We used 16 true and 16 false headlines to create 32 emotionally-charged, fear or anger evoking, variants and 32 emotionally-neutral variants by manipulating the words in the headlines. We also utilized an emotion inoculation video for the treatment condition and an educational video on curling, bananas, or freezer burn for the control condition. Full materials and data can be found on [OSF](#); the inoculation video is available at <https://inoculation.science/>.

Procedure

Participants completed the first attention check in the beginning of the study and then proceeded to provide their consent, which was followed by the second attention check question. The second attention check question was an eliminating question and participants who failed this check were redirected to the end of the study. Participants then answered two questions on their social media usage: the type of content they would consider sharing on social media (Political news, Sports news, Business news, etc), and the social media platforms that they use (Facebook, Twitter, Instagram, etc). Participants who indicated that they do not use any social media platform were redirected to the end of study as well. Since the study entailed watching a video, we next asked participants if the volume on their device was on. For participants who answered in the negative, we provided an additional opportunity to turn the volume on. Those who could not or indicated being unwilling to do so were redirected to the end of the study.

Participants were then randomly assigned to either the treatment or the control condition. Participants in the treatment condition viewed an emotion inoculation video, whereas participants in the control condition viewed an educational video of a similar length on one of three topics: curling, bananas, and freezer burn. The videos were set up such that they would autoplay and were not interactable. Therefore, participants could not pause the videos once they automatically started playing. Moreover, the next button was not made visible till the video was done playing.

Participants then read the instructions about viewing and rating 32 social media headlines. A total of 64 social media headlines generated for this study were evenly split between two blocks. The first block had 16 false and 16 true headlines, eight emotionally-charged and eight emotionally-neutral headlines for each veracity. The second block had emotionally opposite counterparts of the headlines in the first block. Participants only saw one version (high v low emotionality) of each headline and this presentation was counterbalanced across participants. After viewing the randomly-selected headlines, participants rated the trustworthiness and the

manipulativeness of the headline on a 6-point scale: 1) Extremely untrustworthy to 6) Extremely trustworthy, and 1) Not at all manipulative to 6) Extremely manipulative.

All participants completed the third attention check question after rating the headlines. Participants then completed two measures outside the scope of the present investigation. Namely, the actively open-minded thinking scale (AOT, Newton et al., 2023) and the emotion regulation questionnaire (ERQ, Spaapen et al., 2014). Participants then completed the last attention check question, followed by questions related to their political position, and their political preference in general, on social issues, and on economic issues. Finally, participants answered demographics questions which entailed measurement of age, gender, education, income, and ethnicity. Participants also indicated how much they believe in God(s). These measures are also outside the scope of the present investigation and we do not report the results.

We note that a first version of this experiment was run using participants recruited from Lucid (Coppock & McClellan, 2019). In the Lucid experiment, participants were asked either about manipulateness and trustworthiness (as here), or accuracy (as in Experiment 2), or social media sharing intentions. However, there were issues with inattentiveness (30.7% of participants who passed the initial A/V check failed at least one trivial attention check in the Lucid experiment, compared to only 5.6% in the MTurk experiment) and significantly more participants dropped out during the inoculation video compared to the control videos ($\chi^2(1) = 5.26, p = 0.023$) which violates random assignment and undermines causal inference. Thus, we do not consider the results of the Lucid experiment to provide useful insight (although for completeness, we report them in the SI).

Our preregistration (see [OSF](#)) included political salience as a factor, but we report the analysis without this factor included because it is not central to the research question, and including it leads to an overly complicated model with 4-way interactions.

Results

We ran two linear regression models with robust standard errors clustered on subject and headline, predicting (a) manipulateness and (b) trustworthiness using an inoculation treatment dummy (0=control, 1=treatment), a news headline veracity dummy (-0.5=false, 0.5=true), an emotional manipulateness dummy (-0.5=not emotional, 0.5=emotional), and their interactions (Table 2). First, supporting the validity of our items, true content was judged as less manipulative ($b = -1.24 [-1.60, -0.88], z = -6.79, p < .001$) and more trustworthy ($b = 1.46 [1.16, 1.76], z = 9.66, p < .001$) than false content in the control condition, and high emotion content was judged as more manipulative ($b = 0.46 [0.35, 0.57], z = 8.10, p < .001$) and less trustworthy ($b = -0.28 [-0.38, -0.18], z = -5.35, p < .001$) than low emotion content in the control condition - although judgments of manipulateness/trustworthiness were apparently much more influenced by truth than emotionality. Second, replicating Roozenbeek et al. (2022), we also find that the inoculation video significantly increased the difference in manipulateness and trustworthiness judgments

for low versus high emotion content - that is, the treatment increased emotion discernment (manipulativeness: $b = 0.42$ [0.31, 0.53], $z = 7.75$, $p < .001$; trustworthiness: $b = -0.22$ [-0.27, -0.17], $z = -8.50$, $p < .001$); see Figure 2. In particular, the inoculation video significantly increased perceived manipulativenness of high emotion headlines ($b = 0.18$ [0.07, 0.29], $z = 3.09$, $p = .002$, $d = 0.10$) but did not significantly decrease trustworthiness of high emotion headlines ($b = -0.06$ [-0.13, 0.02], $z = -1.50$, $p = .134$, $d = -0.04$). The inoculation also significantly decreased perceived manipulativenness of low emotion headlines ($b = -0.24$ [-0.35, -0.14], $z = -4.48$, $p < .001$, $d = -.14$) and increased trustworthiness of low emotion headlines ($b = 0.16$ [0.09, 0.24], $z = 4.19$, $p < .001$, $d = .10$).

We next ask whether there was a carry-over effect of the inoculation to news headline *truth* discernment; i.e., did the inoculation treatment interact significantly with veracity (either directly or in a three-way interaction with emotional manipulativenness)? As is evident from Table 2, the answer in both cases is no. That is, the inoculation did not interact with news headline veracity for either manipulativenness ($b = -0.07$ [-0.20, 0.07], $z = -0.97$, $p = .332$, $BF < .01$) or trustworthiness ($b = 0.03$ [-0.06, 0.13], $z = 0.69$, $p = .488$, $BF < .01$), nor was there a three-way interaction between the inoculation treatment, headline veracity, and emotionality for manipulativenness ($b = 0.11$ [-0.10, 0.31], $z = 1.03$, $p = .303$, $BF < .01$) or trustworthiness ($b = -0.04$ [-0.14, 0.05], $z = -0.89$, $p = .372$, $BF < .01$). Thus, although the inoculation facilitated the detection of emotional manipulativenness in news headlines, this effect was equivalent for true and false headlines. Furthermore, teaching people about emotional manipulativenness does not (by itself) have an impact on whether they can distinguish between true and false news headlines. Also, as noted, even judgments of manipulativenness were influenced more than twice as strongly by truth than emotionality.

Table 2. OLS regression analyses (b) predicting (a) manipulativenness and (b) trustworthiness using an inoculation treatment dummy (0=control, 1=treatment), a centered news headline veracity dummy (-0.5=false, 0.5=true), an emotional manipulativenness dummy (-0.5=not emotional, 0.5=emotional), and their interactions. Standard errors (clustered on participants and headlines) are in parentheses.

	(a) Manipulativenness	(b) Trustworthiness
(Intercept)	3.682*** (0.096)	3.009*** (0.078)
inoculation treatment	-0.032 (0.051)	0.051 (0.037)
headline veracity (true), centered	-1.243*** (0.183)	1.461*** (0.151)
emotional manipulativenness, centered	0.461*** (0.057)	-0.280*** (0.052)

inoculation * veracity	-0.065 (0.067)	0.034 (0.050)
inoculation * emotionality	0.421*** (0.054)	-0.218*** (0.026)
veracity * emotionality	-0.078 (0.116)	0.121 (0.106)
inoculation * veracity * emotionality	0.105 (0.102)	-0.045 (0.050)
Observations	32,470	32,470
R ²	0.173	0.237
<i>Note.</i> * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$		

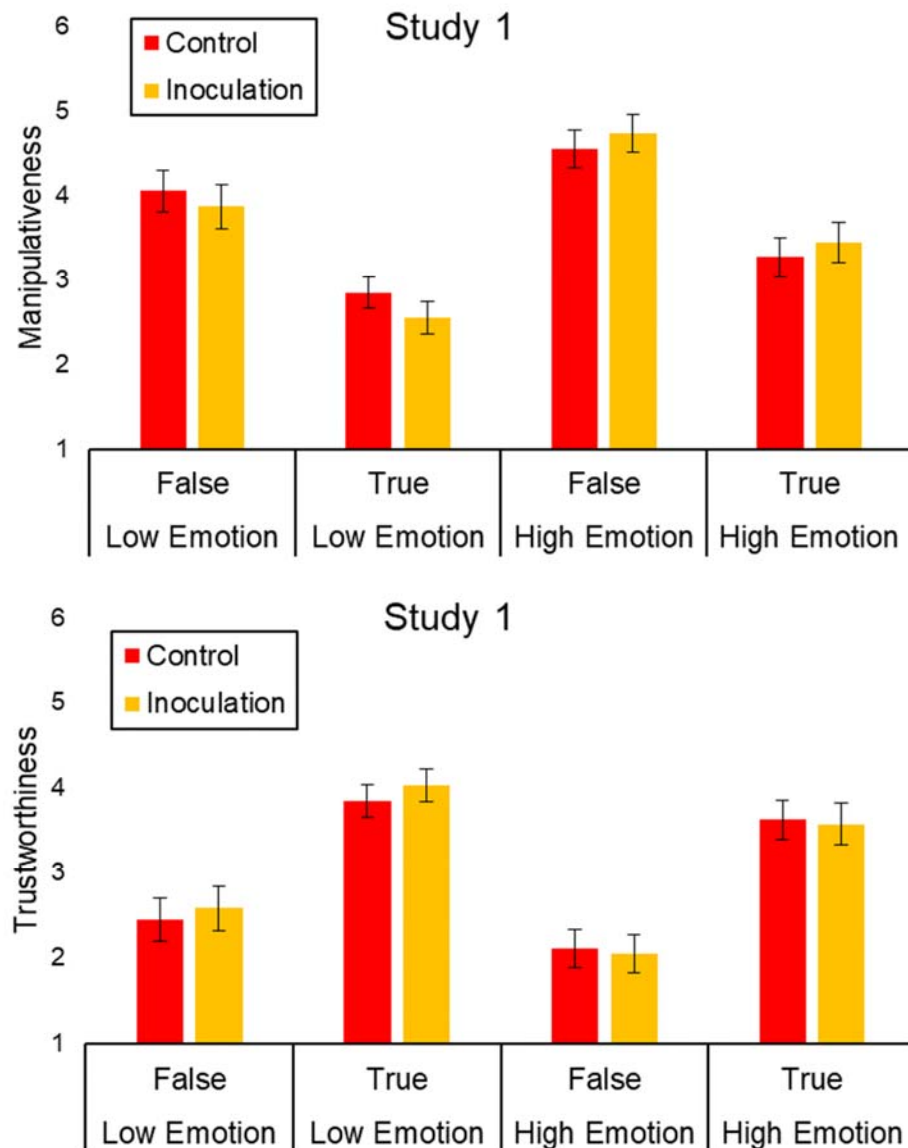


Figure 2. Manipulativenness (left) and trustworthiness (right) ratings by condition and item type in Experiment 1. Error bars indicate 95% confidence intervals, based on robust standard errors clustered on subject and item.

Experiment 2

Experiment 1 confirmed that the inoculation video successfully increased participants' ability to spot emotional manipulation in real-world true and false content. Although this effect was small (d 's = .04-.14), the video inoculation is a relatively short intervention and it did not require any explicit interaction from the participants.

There were, however, no effects on truth discernment *per se*; that is, the treatment did not increase the ability of participants to spot misinformation. Because we manipulated truth and emotionality orthogonally, we were able to isolate the effect of the inoculation, which appears to teach users to identify the emotionality technique without this having an additional carry-over effect to detecting misinformation. One issue, however, is that the identification of the manipulateness technique in Experiment 1 was not “spontaneous” identification in the sense that participants were given the explicit task of searching for manipulateness and trustworthiness. This may have distracted users from focusing on spotting misinformation and therefore may have blunted any carry-over effect of the inoculation on truth discernment. Thus, in Experiment 2, we replicated the exact same design as in Experiment 1, but instead asked participants to judge the accuracy of the claims contained in the news headlines (and, therefore, not manipulateness *per se*). This contributes valuable understanding of how inoculating against a manipulative technique carries over to accuracy judgments, or not.

Method

Participants

Participants were again recruited from Amazon’s Mechanical Turk via Cloudresearch. In total, 2,471 participants entered the experiment and were randomly assigned to rate the accuracy of the headlines or to indicate whether they would share them online. However, 59 participants indicated not having social media and did not therefore qualify for the study and 10 participants failed an attention check at the beginning of the study and did not continue. Finally, 5 participants were removed from the survey because they indicated not being able to turn on their volume and 47 participants quit the survey prior to the intervention. A further 317 participants dropped out and did not contribute data to the primary task. The fraction of participants dropping out did not vary significantly across conditions (14.3% in control, 12.6% in treatment; $\chi^2(1) = 1.50, p = .221$). We therefore had a final sample size of 2,033 participants ($M_{age} = 40$), of whom 793 were male, 1163 were female, 36 chose some other response (e.g., trans/non-binary), and 41 did not respond to the gender question.

Materials

The materials were identical to Experiment 1.

Procedure

Unlike Experiment 1, this study had a 2 (condition: treatment vs. control) x 2 (question asked: accuracy vs. sharing) design. Instead of rating trustworthiness and manipulateness, participants were randomly assigned to either rate the accuracy of the headline (using a scale of 1 = extremely inaccurate to 6 = extremely accurate) or indicate how likely they would be to share the

headline on social media (using a scale of 1 = extremely unlikely to 6 = extremely likely). The procedure of this study was identical to Experiment 1 in all other aspects.

We focus here on the results for the accuracy condition. The results for the sharing condition are presented in the SI. In short, the inoculation did not increase either emotion or truth discernment for sharing⁵.

Results

We ran a linear regression model with robust standard errors clustered on subject and headline, predicting accuracy judgments using an inoculation treatment dummy (0=control, 1=treatment), a news headline veracity dummy (-0.5=false, 0.5=true), an emotional manipulateness dummy (-0.5=not emotional, 0.5=emotional), and their interactions (Table 3). Similar to Experiment 1, true content was judged as more accurate than false content in the control condition ($b = 0.38$ [0.32, 0.44], $z = 11.93$, $p < .001$), and high emotion content was judged as less accurate than low emotion content in the control condition ($b = -0.05$ [-0.07, -0.03], $z = -4.83$, $p < .001$). Second, the inoculation video significantly increased the difference in perceived accuracy for low versus high emotion content - that is, the treatment increased emotion discernment ($b = -0.02$ [-0.03, -0.01], $z = -3.20$, $p = .001$). However, as in Experiment 1, this effect was fairly small (see Figure 3). In particular, the inoculation video significantly decreased perceived accuracy of high emotion headlines ($b = -0.01$ [-0.03, 0.00], $z = -1.97$, $p = .049$, $d = -.04$) but had no significant effect on perceived accuracy of low emotion headlines ($b = 0.01$ [-0.01, 0.02], $z = 0.94$, $p = .345$, $d = .02$).

⁵ Note that the preregistration for Experiment 1 mentions the Experiment 2 dependent variables. We ran Experiment 1 twelve days prior to Experiment 2 (to ensure successful technique recognition), but preregistered them together as one Experiment with 3 dependent variables. We report them separately here for simplicity.

Table 3. OLS regression analysis (b) predicting accuracy judgments using an inoculation treatment dummy (0=control, 1=treatment), a centered news headline veracity dummy (-0.5=false, 0.5=true), a centered emotional manipulativeness dummy (-0.5=not emotional, 0.5=emotional), and their interactions. Standard errors (clustered on participants and headlines) are in parentheses.

	Perceived accuracy
(Intercept)	0.437*** (0.016)
inoculation treatment	-0.004 (0.006)
headline veracity (true), centered	0.380*** (0.032)
emotional manipulativeness, centered	-0.049*** (0.010)
inoculation * veracity	-0.008 (0.010)
inoculation * emotionality	-0.019** (0.006)
veracity * emotionality	0.043* (0.021)
inoculation * veracity * emotionality	-0.001 (0.012)
Observations	32,058
R ²	0.315

Note. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

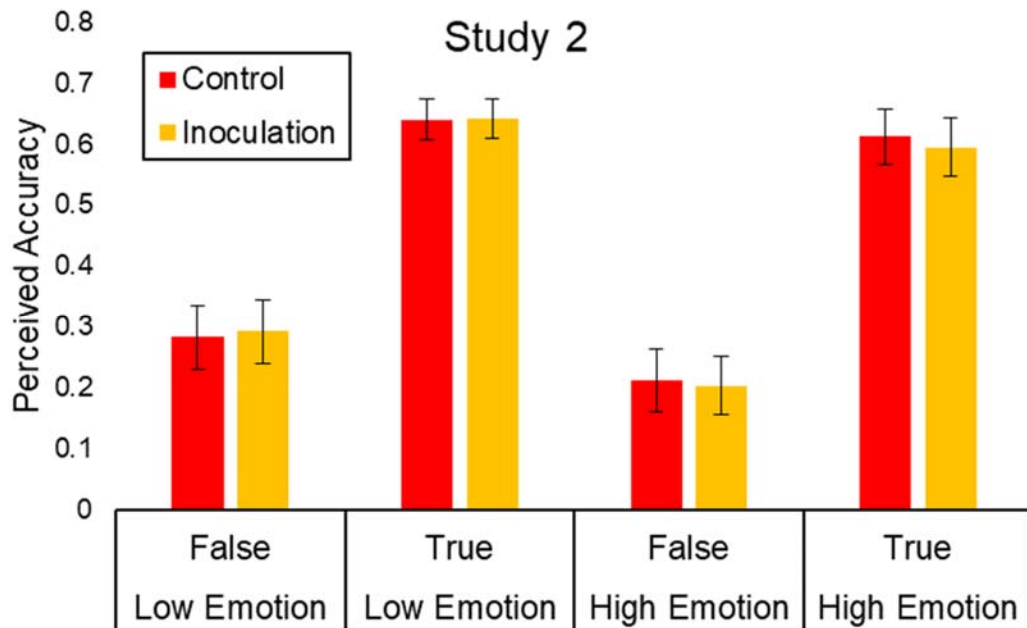


Figure 3. Perceived accuracy by condition and item type in Experiment 2. Error bars indicate 95% confidence intervals, based on robust standard errors clustered on subject and item.

The inoculation treatment did not, however, have a carry-over effect on truth discernment (Table 3). Specifically, there was no interaction between the inoculation and news headline veracity ($b = -0.01 [-0.03, 0.01]$, $z = -0.76$, $p = .446$, $BF = .007$), nor was there a three-way interaction between the inoculation treatment, headline veracity, and emotionality ($b = -.001 [-0.02, 0.02]$, $z = -0.10$, $p = .919$, $BF = .006$).

As a follow-up, we also restricted the analysis to only false-high emotion and true-low emotion items; this represents a maximally favorable limiting case where emotional manipulation is always present among false content and never present among true content. Thus, in this case, decreasing belief in emotional content should be expected to also decrease belief in false content (and therefore increase discernment). However, even here, we still did not find a significant effect of the inoculation on truth discernment (interaction between condition and veracity, $b = 0.01 [-0.01, 0.04]$, $z = 0.83$, $p = .406$, $BF = 0.011$). Accordingly, there was no significant inoculation effect on accuracy judgments for either false-high emotion ($b = -0.01 [-0.03, 0.01]$, $z = -0.84$, $p = .401$, $d = -0.03$, $BF = 0.016$) or true-low emotion ($b = 0.01 [-0.01, 0.02]$, $z = 0.23$, $p = .815$, $d = 0.01$, $BF = 0.011$) headlines.

Experiment 3

Experiment 2 confirmed that the inoculation video decreased belief in emotionally manipulative content relative to non-emotionally manipulative content. Thus, the effects of Experiment 1 (and that of Roozenbeek et al., 2022) do not appear to be driven entirely by explicitly focusing participants on the task of identifying manipulateness (by asking about it directly instead of asking about accuracy, as we did here)⁶. However, as in Experiment 1, there again were no effects on truth discernment. Surprisingly, there was no effect even when only analyzing the subset of items where all false headlines were emotionally manipulative and all true headlines were not. Thus, even if one were to assume that the posited misinformation techniques are *only* present for misinformation (and therefore presenting the most favorable context for testing the effect of the intervention), the inoculation did not apparently help participants spot misinformation.

To test this more directly, we re-ran the experiment but only including the high emotionality false and low emotionality true headlines. This was intended to rule out the possibility that including the fully crossed set of true/false and high/low emotionality headlines undermined the effect of the intervention in some way⁷.

Method

Participants

U.S. participants were recruited from Prolific Academic. In total, 1,526 participants entered the experiment and were randomly assigned to rate the accuracy of the headlines or to indicate whether they would share them online. However, 48 participants indicated not having social media and did not therefore qualify for the study and 7 participants failed an attention check at the beginning of the study and did not continue. Finally, no participants were removed from the survey because they indicated not being able to turn on their volume, but 9 participants quit the survey prior to the intervention. A further 254 participants dropped out and did not contribute data to the primary task. The fraction of participants dropping out did not vary significantly across conditions (17.2% in control, 17.6% in treatment; $\chi^2(1) = 0.04$, $p = .841$). We therefore had a final sample size of 1,208 participants ($M_{age} = 40$), of whom 652 were male, 517 were female, 28 chose some other response (e.g., trans/non-binary), and 11 did not respond to the gender question.

⁶ Of course, participants were nonetheless asked directly about accuracy – see SI for evidence that the inoculation videos did not have the intended effect when participants were asked about sharing the content online.

⁷ We ran a parallel experiment using participants from Mechanical Turk ($N = 1,023$) that is not reported in the SI. We similarly did not find an effect of the emotional inoculation ($b = -0.003$, $p = .734$); however, we believed that the data were questionable because the difference between true and false headlines, which is robustly observed to be large in all other studies in this paper as well as in the literature more generally, was extremely small ($b = .04$, relative to $b = .48$ in the parallel Experiment 3). Thus we do not consider this study further.

Materials and Procedure

Out of the 32 emotionally-charged (16 true and 16 false) and 32 emotionally-neutral (16 true and 16 false) headlines used in Experiment 1, we kept only the 16 false emotionally-charged headlines and the 16 true emotionally-neutral headlines for use in this study.

Participants provided accuracy ratings as in Experiment 2 (there was no sharing condition). The 16 headlines seen by the participants came from one of two randomly-chosen blocks. The first block had eight false emotionally-charged and eight true emotionally-neutral headlines. The second block had a different set of false emotionally-charged and true emotionally-neutral headlines, eight of each kind. The remaining procedure was identical to Experiment 2, except that we removed the AOT, ERQ, and the fourth attention check (owing to the shorter length).

The analyses were preregistered (see [OSF](#)).

Results

In this experiment, emotional manipulateness and headline truth were manipulated to be fully consistent with each other. Thus, we ran a linear regression model with robust standard errors clustered on subject and headline, predicting accuracy judgments using an inoculation treatment dummy (0=control, 1=treatment), a news headline veracity/emotional manipulateness dummy (0=false/high emotional, 1=true/low emotional), and their interaction (Table 4). As expected, true low emotional content was judged as more accurate than false high emotional content in the control condition ($b = 0.48$ [0.42, 0.54], $z = 15.48$, $p < .001$). However, the inoculation video did *not* significantly increase the difference in perceived accuracy for true low emotion versus false high emotion content ($b = -0.01$ [-0.03, 0.02], $z = -0.58$, $p = .559$, $BF = .009$). That is, participants did not improve in their ability to distinguish between true and false news headlines even though we manipulated the true headlines to be low emotion and the false headlines to be high emotion. There was no significant inoculation effect on accuracy judgments for either false-high emotion ($b = 0.003$ [-0.01, 0.02], $z = 0.40$, $p = .691$, $d = -0.01$, $BF = 0.011$) or true-low emotion ($b = -0.004$ [-0.02, 0.02], $z = -0.37$, $p = .709$, $d = 0.01$, $BF = 0.011$) headlines (see Figure 4).

Table 4. OLS regression analysis (b) predicting accuracy judgments using an inoculation treatment dummy (0=control, 1=treatment), a news headline veracity/emotional manipulativenness dummy (0=false/high emotional, 1=true/low emotional), and their interaction. Standard errors (clustered on participants and headlines) are in parentheses.

	Perceived accuracy
(Intercept)	0.163*** (0.024)
inoculation treatment	0.003 (0.009)
headline manipulativenness	0.479*** (0.031)
inoculation * manipulativenness	-0.007 (0.012)
Observations	19,266
R ²	0.446

Note. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

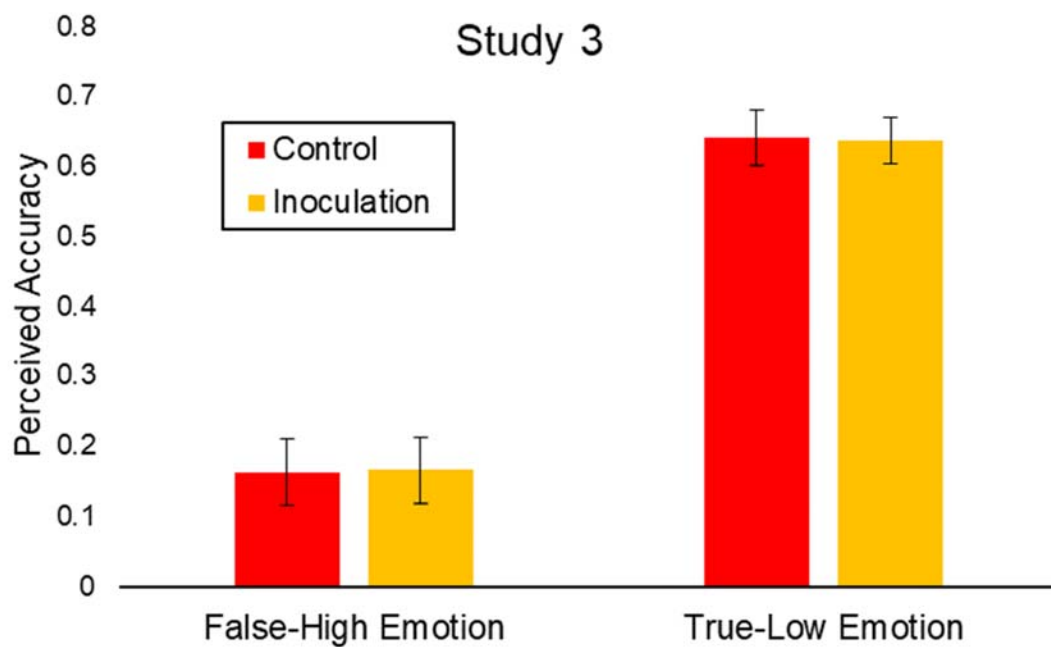


Figure 4. Perceived accuracy by condition and item type in Experiment 3. Error bars indicate 95% confidence intervals, based on robust standard errors clustered on subject and item.

Experiment 4

Experiment 3 set a highly favorable context for testing the effect of the inoculation on discerning between true and false news headlines – i.e., participants only received false headlines that contained the emotional manipulation technique and true headlines that did not contain the technique; hence, identifying the technique should correspond directly to detecting misinformation. Notwithstanding, the inoculation did not produce a significant difference between true (low emotion) and false (high emotion) videos. Thus, the emotional language inoculation, at least in the present form, appears to be very specific: It helps people identify emotionality in a general sense (as evidenced in Experiments 1 and 2), but its effect is undermined by some combination of (a) asking about accuracy (instead of the presence of the technique *per se*) and (b) the presence of claims that are either true or false (rather than the techniques presented on their own without claims of fact). That is, when the task is made more difficult by intermixing actual true or false claims, the video appears to lose its effectiveness as an “inoculation against misinformation”.

One potential reason for this is that the emotional language inoculation video does not actually draw attention to accuracy in any specific way. Rather, the video focuses on identifying emotionality techniques as a form of grabbing one’s attention. Although the videos were proposed as a misinformation intervention, it may be that they do not actually draw people’s attention to whether content is true or false. Other research has shown that, at least in terms of decisions about what to share online, users may often fail to even consider whether content is accurate before they decide to share that content (Pennycook et al., 2021). Indeed, subtly reminding people about accuracy is sufficient to increase the quality of content (i.e., less false relative to true content) that people are willing to share online (Pennycook & Rand, 2022). One possibility, then, is that a simple reminder about the importance of considering accuracy (and the threat of misinformation) prior to the inoculation video may help users apply the specific content that they learn about emotional manipulation in the video. In support of this possibility, previous inoculation research that used longer videos with a broader range of techniques and that also included explicit reference to accuracy has found an effect on judgments of the reliability (a proxy for accuracy) of subsequent misinforming material (Lewandowsky & Yesilada, 2021) (albeit, as discussed, material that does not contain true or false claims *per se*). To test this possibility, we modified the inoculation video so that it was bookended with reminders about accuracy (video can be viewed at [OSF](#)).

Method

Participants

U.S. participants were recruited from Prolific Academic. In total, 1,348 participants entered the experiment and were randomly assigned to rate the accuracy of the headlines or to indicate whether they would share them online. However, 27 participants indicated not having social

media and did not therefore qualify for the study and 4 participants failed an attention check at the beginning of the study and did not continue. Finally, no participants were removed from the survey because of the intervention. A further 95 participants dropped out and did not contribute data to the primary task. The fraction of participants dropping out did not vary significantly across conditions (6.8% in control, 7.8% in treatment; $\chi^2(1) = 0.45, p = .504$). We therefore had a final sample size of 1,211 participants ($M_{age} = 40$), of whom 647 were male, 514 were female, 37 chose some other response (e.g., trans/non-binary), and 13 did not respond to the gender question.

Materials and Procedure

Videos used in this study were modified to include accuracy prompts. This new video had accuracy reminders in the beginning and at the end, with the inoculation video used in the previous experiments ‘sandwiched’ in the middle (see [OSF](#)). The goal was to increase the chances that the viewer would apply the content of the video to the issue of truth versus falsity. The three educational videos in the control condition were also replaced to match the duration of the new treatment video. The new educational control videos were about stars, Scotland, and bats. Headlines and procedure were the same as in Experiment 3. The analyses were preregistered (see [OSF](#)).

Results

As in Experiment 3, emotional manipulateness and headline truth were manipulated to be fully consistent with each other. Thus, we ran a linear regression model with robust standard errors clustered on subject and headline, predicting accuracy judgments using an inoculation (with accuracy prompt) treatment dummy (0=control, 1=treatment), a news headline veracity/emotional manipulateness dummy (0=false/high emotional, 1=true/low emotional), and their interaction (Table 5). True low emotional content was judged as more accurate than false high emotional content in the control condition ($b = 0.45 [0.39, 0.51], z = 14.99, p < .001$). Critically, the inoculation video *did* significantly increase the difference in perceived accuracy for true low emotion versus false high emotion content ($b = 0.03 [0.01, 0.06], z = 2.95, p = .003, d = 0.16, 7.6\%$ increase). That is, the combined inoculation and accuracy prompt video was successful in leading participants to improve in their ability to distinguish between true and false news headlines (when true headlines were manipulated to be low emotion and false headlines were manipulated to be high emotion). Interestingly, the inoculation did not have a significant effect on accuracy judgments for false-high emotion headlines ($b = -0.01 [-0.03, 0.01], z = -1.08, p = .282, d = -0.036$) but significantly increased accuracy judgments for true-low emotion headlines ($b = 0.02 [0.01, 0.04], z = 2.79, p = .005, d = 0.089$) (see Figure 5).

Table 5. OLS regression analyses (b) predicting accuracy judgments using a combined inoculation and accuracy prompt treatment dummy (0=control, 1=treatment), a news headline veracity/emotional manipulateness dummy (0=false/high emotional, 1=true/low emotional), and their interaction. Standard errors (clustered on participants and headlines) are in parentheses.

	<u>Perceived accuracy</u>
(Intercept)	0.169*** (0.022)
inoculation & accuracy prompt treatment	-0.009 (0.008)
headline manipulateness	0.448*** (0.030)
inoculation & accuracy prompt * manipulateness	0.034** (0.011)
Observations	19,336
R ²	0.438

Note. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

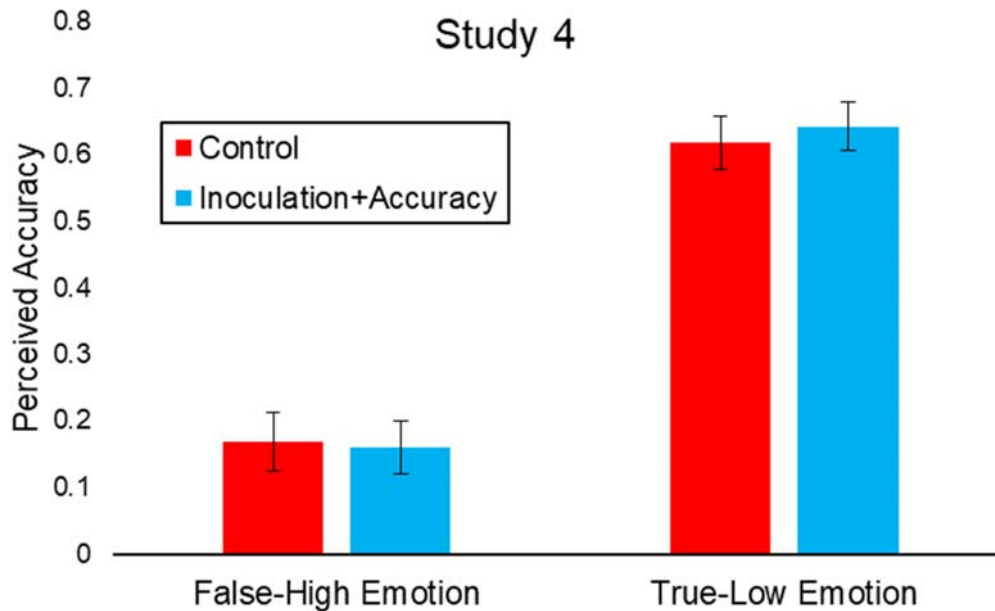


Figure 5. Perceived accuracy by condition and item type in Experiment 4. Error bars indicate 95% confidence intervals, based on robust standard errors clustered on subject and item.

Experiment 5

Experiment 4 provided evidence that the inoculation was effective at increasing truth discernment when paired with an accuracy prompt. Although this effect was small and evident only for true-low emotion headlines (as opposed to false-high emotion headlines), it nonetheless provides some evidence that participants were able to apply what they learned about emotional manipulation in the short video to content that contains true and false claims. Nonetheless, one question remains: Was the accuracy prompt itself responsible for the increase in truth discernment, or are both elements necessary to produce the effect? Based on past theorizing (Pennycook & Rand, 2022), it is unlikely that the accuracy prompt alone would improve judgments of accuracy since the intervention simply draws attention to accuracy without teaching participants any new information. Furthermore, our participants are already being directly asked about accuracy as our central outcome measure – and, thus, the task is already directly drawing their attention to accuracy (asking accuracy as an outcome has been shown to be a maximally strong accuracy prompt; Pennycook et al., 2021). Nonetheless, we used a novel accuracy prompt in this study and we cannot rule out the possibility that it had some sort of carry-over effect on its own (e.g. if it was doing more than just cueing accuracy, such as applying normative pressure or increasing motivations to be accurate). Thus, we test for carry-overs from the accuracy prompt in Experiment 5 by comparing the combined accuracy prompt and inoculation video with the accuracy prompt video on its own.

Method

Participants

U.S. participants were recruited from Prolific Academic. In total, 1,996 participants entered the experiment and were randomly assigned to rate the accuracy of the headlines or to indicate whether they would share them online. However, 62 participants indicated not having social media and did not therefore qualify for the study and 4 participants failed an attention check at the beginning of the study and did not continue. Finally, 1 participant was removed from the survey because they indicated not being able to turn on their volume, and 11 participants quit the survey prior to the volume question. A further 114 participants dropped out and did not contribute data to the primary task. The fraction of participants dropping out did not vary significantly across conditions (6.0% in long control, 6.6% in combined inoculation and accuracy, 5.5% in short control, 5.8% in accuracy only; $\chi^2(3) = 0.59, p = .899$). We therefore had a final sample size of 1,804 participants ($M_{age} = 40$), of whom 907 were male, 838 were female, 49 chose some other response (e.g., trans/non-binary), and 11 did not respond to the gender question.

Materials and Procedure.

In this study, we had two sets of videos. As in Experiment 4, we had a ‘sandwich’ inoculation video with accuracy reminders in the beginning and at the end for the treatment condition, and three duration-matched educational videos (on stars, Scotland, and bats) for the control condition. Unlike Experiment 4, we additionally created a condition where participants only received the initial accuracy-prompt component of the video. Consequently, we used a different set of three duration-matched educational/motivational videos (on recycling, entrepreneurship, and effort) for the control condition. The materials and procedure were otherwise identical to Experiment 4. The analyses were preregistered (see [OSF](#)).

Results

As preregistered, we first confirmed that there were no differences between the two sets of control videos (condition dummy: $b = -0.01$ $[-0.03, 0.01]$, $z = -0.90$, $p = .366$, $BF = 0.014$; interaction between condition and veracity: $b = 0.003$ $[-0.02, 0.03]$, $z = 0.24$, $p = .807$, $BF = 0.007$). Thus, we pooled the control conditions. We ran a linear regression model with robust standard errors clustered on subject and headline, predicting accuracy judgments using a combined inoculation and accuracy prompt treatment dummy (0=control, 1=treatment), an only accuracy prompt treatment dummy (0=control, 1=treatment), a news headline veracity/emotional manipulateness dummy (0=false/high emotional, 1=true/low emotional), and the interactions between manipulateness and our two treatment conditions (Table 6). True low emotional content was judged as more accurate than false high emotional content in the control ($b = 0.45$ $[0.39, 0.51]$, $z = 14.34$, $p < .001$).

Critically, replicating Experiment 4, the inoculation video significantly increased the difference in perceived accuracy for true low emotion versus false high emotion content ($b = 0.05$ $[0.02, 0.07]$, $z = 3.40$, $p = .001$, $d = 0.22$, 10.2% increase). In this case, the combined inoculation and accuracy prompt video did not significantly decrease perceptions of accuracy for false-high emotion headlines ($b = -0.03$ $[-0.06, 0.00]$, $z = -1.88$, $p = .060$, $d = -0.088$) but significantly increased accuracy judgments for true-low emotion headlines ($b = 0.02$ $[0.00, 0.04]$, $z = 2.14$, $p = .033$, $d = 0.090$) (see Figure 6). As expected, there was no interaction effect for the accuracy prompt alone ($b = 0.001$ $[-0.02, 0.02]$, $z = 0.07$, $p = .948$, $BF = .009$), indicating that simply drawing people’s attention to the problem of misinformation is not sufficient to increase subsequent truth discernment.

Importantly, the effect of the combined treatment on discernment was significantly larger than the effect of the accuracy prompt alone (Wald test comparing the coefficient on (i) the interaction between accuracy prompt only and veracity with (ii) the interaction between combined inoculation+accuracy and veracity, $\chi^2(1) = 11.33$, $p < .001$). In other words, the accuracy prompt and inoculation treatment had a synergistic effect on truth discernment, but similar to prior experiments, neither were effective on their own. Interestingly, since prior work on accuracy prompts has focused on news headline sharing (and not belief, as we measure here),

this research also indicates that inoculations may boost the effect of prompting accuracy beyond choices about what to share.

Table 6. OLS regression analyses (b) predicting accuracy judgments using an inoculation treatment dummy (0=control, 1=treatment), a news headline veracity/emotional manipulateness dummy (0=false/high emotional, 1=true/low emotional), and their interaction. Standard errors (clustered on participants and headlines) are in parentheses.

	<u>Perceived accuracy</u>
(Intercept)	0.175*** (0.025)
inoculation & accuracy prompt treatment	-0.022* (0.011)
accuracy prompt treatment	-0.009 (0.010)
headline manipulateness	0.452*** (0.031)
inoculation & accuracy prompt * manipulateness	0.046** (0.013)
accuracy prompt * manipulateness	0.001 (0.011)
Observations	28,8812
R ²	0.438

Note. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

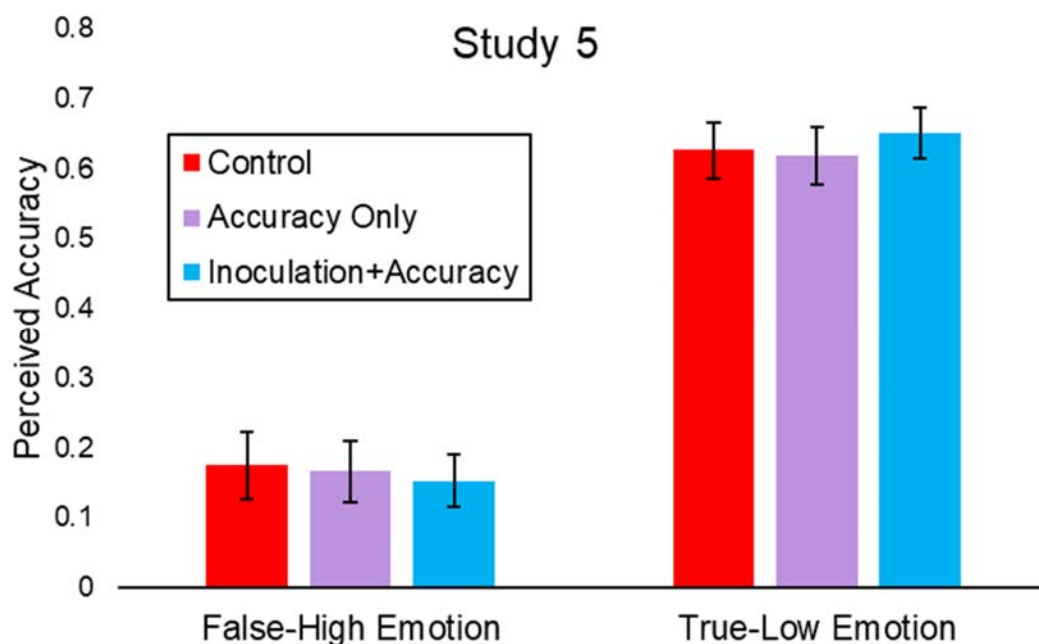


Figure 6. *Perceived accuracy by condition and item type in Experiment 5. Error bars indicate 95% confidence intervals, based on robust standard errors clustered on subject and item.*

General Discussion

A common sentiment is that multiple tools will be needed to effectively combat misinformation (Bak-Coleman et al., 2022; Bode & Vraga, 2021). We present evidence here that popular intervention approaches can have synergistic effects, even when not effective on their own.

In particular, we tested a popular inoculation intervention intended to increase people’s ability to recognize emotional language manipulation in text. Unlike past work, we tested this intervention using stimuli that not only contained the emotional language manipulation (in contrast to neutral/low emotionality content), but that also contained true or false claims - allowing us to assess the effect of the inoculation on truth discernment.

Our results identified a limitation of the emotional inoculation intervention. Consistent with past work (e.g. Roozenbeek et al., 2022), the intervention was successful in helping people distinguish between content that contains (vs. does not contain) an emotional language manipulation. Critically, however, we found consistent evidence that the inoculation on its own did *not* help people distinguish between true and false content. In fact, we did not find an inoculation effect on accuracy discernment even in the extreme case where all false claims - and no true claims - contained emotional manipulation.

Why did the emotional inoculation fail to facilitate truth discernment? One plausible mechanism is that the emotional inoculation that we focused on (which was among the strongest interventions in past work; Roozenbeek et al., 2022) did not draw an explicit connection between the use of emotional language and falsehoods *per se*. This is important because past work shows that people tend to be inattentive to accuracy on social media (Epstein et al., 2023). Consistent with this, we found that focusing people's attention on accuracy prior to watching the inoculation video (as is done in popular accuracy prompt interventions; Pennycook & Rand, 2022) *did* lead to an increase in truth discernment - as a consequence of the combined force of inoculation and accuracy prompting. This demonstrates a synergy between inoculation and accuracy prompt interventions, albeit with modest effect sizes (7.6-10.2% increase in accuracy discernment relative to the control, $d = 0.16-0.22$). Interestingly, however, this increase in truth discernment was primarily the result of increasing belief in true-low emotion claims (as opposed to decreasing belief in false-high emotion claims).

Implications

These findings emphasize the importance of jointly deploying multiple interventions. Building off recent theorizing about boost versus nudge interventions, our results suggest that an inoculation “boost” – i.e., an intervention that increases people's cognitive and motivational competencies through learning (Hertwig & Grüne-Yanoff, 2017) – is important for improving truth discernment *capabilities*, whereas an accuracy prompt nudge is critical for *engaging/activating* that capability.

Our findings have implications for work on psychological inoculation in the context of misinformation. In particular, misinformation inoculation interventions have typically been evaluated by testing their effects on people's ability to identify manipulative techniques. This is usually done using stylized or “synthetic” materials that do not contain actual true or false claims, in an effort to offer “clean” examples of the manipulative technique absent real-world truth claims that might influence people's evaluations.

The logic behind this approach is that helping people recognize common manipulation techniques, such as emotional language manipulation, ought to help them distinguish a wide range of misinformation, which includes misleading content as well as false content. Our findings, however, indicate that simply providing information about manipulative techniques may *not* be sufficient to lead people to be “inoculated” against belief in actual falsehoods. Rather, only when people were prompted to connect the information about emotional manipulation with factuality *per se* did we find evidence that the inoculation had an impact on truth discernment.

The implication of this is that prior inoculation approaches may only be broadly effective. If they have successfully connected the information that people are learning about manipulation techniques with the threat of misinformation. Moreover, even in such cases, past work that has

tested inoculation approaches using technique identification may have created inflated expectations regarding effect sizes. For example, Experiment 1 found effect sizes of $d = 0.41$ and $d = 0.58$ on technique discernment using manipulateness and trustworthiness ratings as the outcome (in line with Roozenbeek et al.'s (2022) finding of $d = 0.49$ on technique discernment). In contrast, however, Experiment 2 found an effect size of only $d = 0.20$ on technique discernment when using identical materials but with perceived accuracy as the outcome— and, as mentioned above, found no significant effect ($d = -0.03$) on truth discernment. Furthermore, when we did ultimately find a significant improvement in truth discernment from the combined inoculation+accuracy prompt intervention in Experiments 4 and 5, this effect was driven by increased belief in low-emotion true headlines more so than decreased belief in high-emotion false headlines (i.e., the effect was significant for low-emotion true headlines in both studies, but non-significant for high-emotion false in Experiment 4 and but failed to reach significance in Experiment 5, $p = .06$). Participants were taught that emotionality should be viewed with skepticism and this led to an increase in perceived accuracy for true content that did *not* contain emotional language. This is similar to the “implied truth effect” wherein putting warning labels on some news may increase belief in unlabeled news ((Pennycook et al., 2020). Nonetheless, it is surprising that the inoculation did not have a stronger negative effect on the high-emotion false claims than a positive effect on low-emotion true claims. Future work should reconcile this pattern with inoculation theory’s focus on reducing susceptibility to manipulative content *per se*.

Limitations

One important empirical limitation of our research is that we found a successful inoculation (with accuracy prompt) effect on truth discernment in a context where the emotional language manipulation was fully confounded with truth. That is, the false content all contained an emotional language manipulation, and the true content did not contain any emotional language manipulation. There is reason to expect false content to be somewhat more likely to contain emotional manipulation than true content. For example, Carrasco-Farré (2022) showed in analysis of nearly 100,000 news articles that fake news articles contain more negative sentiment on average than factual articles. However, emotional language and truth/falsity are certainly far from perfectly correlated in the real world. Thus, to whatever extent a disconnect exists between emotionality and veracity outside the lab setting, the true effect on truth discernment will be smaller than the average effect of $d = 0.20$ observed for the combined intervention in our studies.

Related to this, although our news headlines were sourced from real-world content, our emotional language manipulation was artificial (in the sense that we created the language of the manipulation ourselves). Although this is typical of past work (e.g., Roozenbeek et al., 2022), it is possible that our manipulations were weaker (or stronger) than would be common in real-world content. Although our pretest allowed us to validate that our emotional language manipulation successfully induced emotions (a step not common in past work), this does not

necessarily guard against the possibility that our manipulation was unrepresentative of emotional content in the field.

Another limitation of the present studies is the reliance on headlines alone, which provides a very limited attack surface for the inoculation intervention to find traction. It is therefore possible that a more realistic extent of content (e.g., a headline plus a lede or even an entire news article) will provide more opportunity for the intervention to “grab”. Although this remains an open empirical question, it is quite plausible that inoculation effect sizes may be greater when there is more content that the intervention can be applied to.

Furthermore, this study only tested one medium and format of interventions for both inoculation and accuracy prompts - animated video. It is worth testing variations of videos - live action, creator-driven, humorous - as well as other mediums entirely such as text, audio, or captioned images. Similarly, we used a specific set of 32 headlines. Future work should investigate how the results generalize to other headline sets. We also focused only on emotional language manipulation. Although this is a particularly salient and common misinformation technique (Carrasco-Farré, 2022; Paschen, 2019), it is nonetheless important to investigate whether our results generalize to other misinformation techniques.

Future work should look into the mechanism at play in the accuracy prompt used in our study. It is unclear whether the prompt in our study was priming accuracy (as in past work on sharing intentions) or something else, such as linking the inoculation to misinformation. The portion of the video including the prime did mention accuracy, but accuracy was not the primary focus. Past work has used a variety of accuracy prompt methods (e.g., Epstein et al., 2021; Pennycook & Rand, 2022); however, one limitation is that these tend to be varied and may differ in terms of how closely connected they are to the key mechanism(s) of interest. Future work clarifying this issue would be beneficial.

Conclusion

We conclude that combining interventions can have synergies for helping people better distinguish between true and false information online. Theoretical modeling has suggested the existence of synergies from combining approaches to fight misinformation (Bak-Coleman et al., 2022; Bode & Vraga, 2021). The present studies present empirical evidence of this phenomenon using randomized-controlled trials with participants viewing scalable interventions and real world headlines. Our study has significant implications for the burgeoning field designing misinformation interventions. The practitioners, technologists, and policymakers developing these interventions would benefit from testing and deploying multiple methods to fight misinformation in tandem where they are synergistic. Combined interventions can remain highly scalable; in this case, both interventions were delivered together in a short video of about two minutes.

Notwithstanding the potential success of such scalable interventions, they can only be part of the solution to the misinformation problem. Misinformation does not arise in a vacuum but flourishes as part of an online ecosystem whose architecture focuses on capturing user attention (e.g., Kozyreva et al., 2020). Given that purveyors of misinformation are, by definition, freed from constraints imposed by reality, they can exploit known biases in human attention (e.g., by presenting outraging content (e.g., Bakir & McStay, 2018), which platform algorithms are likely to amplify). Interventions such as those developed here must therefore be accompanied by creating an internet with democratic credentials and based on user empowerment (Lewandowsky & Pomerantsev, 2022).

Acknowledgements

This research was supported by funding from Jigsaw (Google). GP acknowledges financial support from the Social Sciences and Humanities Research Council of Canada, the John Templeton Foundation and the Office of Naval Research. DR and AB acknowledge support from the National Science Foundation (NSF Award 2047152) and the Alfred P. Sloan Foundation (Grant #2021-16891). SL acknowledges financial support from the European Research Council (ERC Advanced Grant 101020961 PRODEMINFO), the Humboldt Foundation through a research award, the Volkswagen Foundation (grant "Reclaiming individual autonomy and democratic discourse online: How to rebalance human and algorithmic decision making"), and the European Commission (Horizon 2020 grant 101094752 SoMe4Dem). SL also receives funding from UK Research and Innovation (through EU Horizon replacement funding grant number 10049415). The views and conclusions contained herein are those of the authors and should not be interpreted as representing the official policies, either expressed or implied, of the funding bodies.

Competing interests

GP and DR have received funding from Meta and Google; GP, AB, and RC were employed at Google during the time that the studies were designed and the data was collected, and BG both was and is employed at Google. Google has invested in both inoculation and accuracy prompts as misinformation interventions.

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